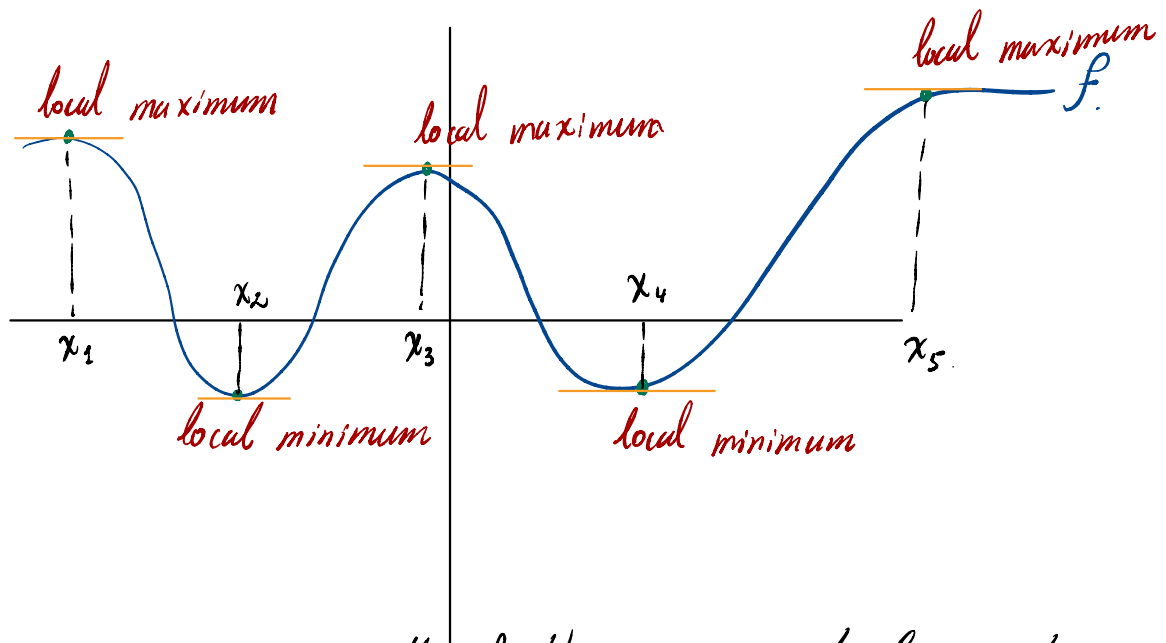


About Math 224: We know that if $f(x)$ is a function:

Dom f , Range f , Critical points, inflection points,
 are points in Dom f s.t. $f'(x_0) = 0$ or $f'(x_0) = \text{DNE}$.
 are points in Dom f s.t. $f''(x_0) = 0$.
 First Derivative Second Derivative.

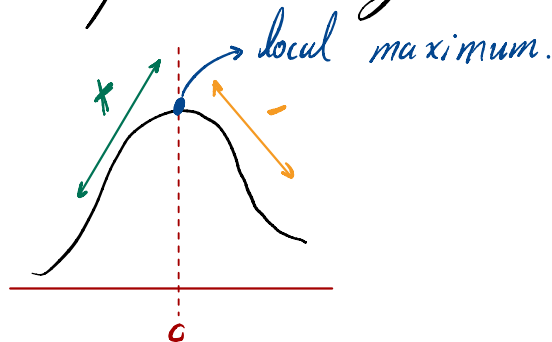


then $x_1, x_2, x_3, x_4, x_5, \dots$ all of them are critical points. Such that $f'(x_i) = 0$ for $i = 1, 2, 3, 4, 5$.

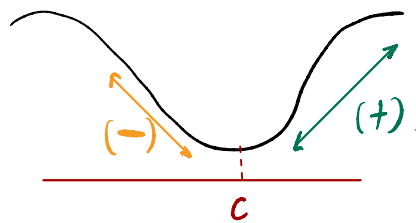
The First Derivative test.

Suppose that c is a critical point of a continuous function f .

a). If f' changes from positive to negative at c , then f has a local maximum at c .



b). If f' changes from negative to positive at c , then f has a local minimum at c .



c). If f' is positive to the left and right of c , or negative to the left and right of c , then f has no local maximum or minimum at c .

Second Derivative test.

Suppose that f'' is continuous near c .

a). If $f'(c)=0$ and $f''(c)>0$, then f has a local minimum at c .

b). If $f'(c)=0$ and $f''(c)<0$, then f has a local maximum at c .